Sushant Lokhande

San Jose, CA | +16693406655 | lokhandesushant014@gmail.com | sushantlokhande.me | linkedin.com/in/sushantlokhande14 | github.com/sushantlokhande14

Professional Summary

Software Systems Engineer with experience building **data-intensive platforms, ML-adjacent tooling, validation pipelines, and full-stack applications** using Python, TypeScript, React, C++, SQL, and APIs. Experienced developing **human-in-the-loop workflows, automated quality checks, data pipelines, observability, and scalable services** supporting high-volume evaluation and autonomy-oriented systems.

Technical Skills

Programming: Python, TypeScript, JavaScript, C++, Go, SQL
Frontend: React, Redux, HTML, CSS, Dashboard Development, State Management
Backend & APIs: FastAPI, Node.js, GraphQL, REST APIs, gRPC, Protobuf, Async Processing
Data & ML Systems: PyTorch, Data Pipelines, Human-in-the-Loop Workflows, Auto-QA, Data Validation, Model Services
Data Stores: PostgreSQL, MySQL, Redis, Data Modeling, Schema Design, Query Optimization
Engineering: OOP, Design Patterns, Data Structures, Algorithms, pytest, CI/CD, Docker, Grafana, Prometheus

Experience

SJSU | Graduate Assistant, Data & Autonomous Systems — Aug 2025 to May 2026

- Reduced validation turnaround by **52%** by building Python and C++ pipelines that compared software builds, processed scenario telemetry, detected behavioral regressions, and generated structured failure reports.
- Improved automated quality detection by **41%** by implementing rule-based checks, anomaly thresholds, metadata validation, and reusable evaluation logic across large experiment and telemetry datasets.
- Accelerated failure investigation by **3.0x** by correlating scenario metadata, SQL records, execution logs, telemetry signals, and software-version differences into reproducible debugging workflows.
- Reduced debugging effort by **44%** by instrumenting pipeline latency, processing failures, scenario outcomes, and system health through Grafana and Prometheus observability.

Turito | Software Engineer Intern — Jun 2022 to Oct 2022

- Reduced API latency by **55%** by profiling Python services, optimizing PostgreSQL queries, introducing Redis caching, and eliminating redundant backend processing.
- Scaled event-processing throughput by **2.4x** by implementing asynchronous workers, batched persistence, queue-based processing, and parallel execution across data-intensive services.
- Improved data quality by **43%** by introducing schema validation, idempotent writes, malformed-input handling, reconciliation jobs, and automated checks before downstream processing.
- Reduced release regressions by **38%** by expanding pytest coverage, adding integration tests, strengthening code reviews, and automating deployment validation through CI/CD.

Trivia Softwares | Software Engineering Intern — Nov 2021 to Jan 2022

- Built backend data services across **500K+ records** by implementing Python APIs, SQL persistence, validation logic, and reusable data-processing components.
- Reduced data-processing latency by **39%** by vectorizing transformations, batching operations, caching reusable results, and optimizing database access patterns.
- Improved model-serving reliability by **31%** by adding input validation, structured error handling, inference checks, and regression tests around ML-backed API workflows.
- Expanded automated validation coverage by **32%** by developing pytest suites, integration tests, malformed-input scenarios, and reproducible Docker environments.

Relevant Projects

Human-in-the-Loop Evaluation Platform | *Python, React, FastAPI, PostgreSQL, PyTorch*

- Automated evaluation for **500+ submissions** by building parallel workflows for data ingestion, automated assessment, code execution, error detection, human review, and structured feedback.
- Reduced manual review effort by **46%** by introducing confidence-based routing, automated quality checks, reusable validation rules, and reviewer-facing diagnostics for ambiguous outputs.

Autonomous Vehicle Validation Platform | *Python, C++, SQL, Grafana, Prometheus*

- Reduced regression triage time by **58%** by building scenario-comparison pipelines that aligned telemetry, software versions, expected behaviors, and failure classifications across repeated simulations.
- Improved experiment reproducibility by **42%** by standardizing scenario metadata, validation rules, dataset versions, execution artifacts, and automated reporting for downstream analysis.

Education

San Jose State University | M.S. in Computer Science — May 2026
University of Mumbai | B.E. in Computer Engineering — May 2023